CS341 · COMPUTER GRAPHICS

Chapter 1 Overview of Graphics Systems

Midterm summary

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What's inside

01 Two kinds of systems: random and raster

02 Random scan: advantages and disadvantages

03 Raster: the frame buffer and memory

04 Input and hard-copy devices

05 Display devices

06 The CRT and its parts

07 Persistence and the refresh CRT

08 Resolution and aspect ratio

09 Flat-panel displays

10 Screen clarity

11 Video card and graphics software

01 Two kinds of systems: random and raster

1 Random Scan

The electron beam draws the picture directly, line by line, like a pen. The picture is stored as line-drawing commands.

Also called **vector, stroke-writing, or calligraphic**. The electron beam **directly draws** the picture.
Picture definition is stored as a set of line-drawing commands in the **display list** (refresh display file, vector file, or display program).
Pen plotter and laser light show are also random scan devices.

2 Raster Scan

The beam sweeps the whole screen one row at a time from top to bottom (TV technology) and turns each dot on or off. The picture is stored as pixel values in the frame buffer.

Scans the screen **from top to bottom** in a regular pattern, one row at a time. Each row is a **scan line**.
Screen point = **pixel** (picture element).
Picture information is stored in the **refresh (frame) buffer**.

	Random Scan	Raster Scan
Drawing	The beam draws lines directly	Sweeps the screen line by line
Storage	**Display list**	**Frame buffer (bit map)**
Resolution	Very high, smooth lines	Stair-step lines
Color and shading	Limited; cannot display realistic shaded scenes	Easy
Examples	Pen plotter, laser light show	PC monitors, TV, printers (dot matrix and laser)

02 Random scan: advantages and disadvantages

Advantages

Very high resolution, limited only by monitor
Easy animation: just draw at different positions
Requires little memory (just enough to hold the display program)

Disadvantages

Requires an intelligent (processor-controlled) electron beam
Limited screen density before flicker
Can't draw a complex image
Limited color capability (very expensive)
Cannot display realistic shaded scenes

03 Raster: the frame buffer and memory

How does the beam know when to turn on and off? The pattern is stored in a special graphics memory called the frame buffer (or bit map), where each location corresponds to a pixel. The display processor scans the memory and turns the beam on for 1 and off for 0.

- Interlaced scan: the first pass draws the even lines 0, 2, 4, ... and the second the odd lines 1, 3, 5, ...
- It must run above 45 Hz or the screen flickers.
- The video controller reads x and y coordinates and the pixel value from the frame buffer and outputs the x deflection, y deflection, intensity and color.

Memory size

```
Memory = width × height × bits per pixel
```

```
Monochrome 640 × 480:  307,200 bits ÷ 8 = 38,400 bytes
```

In the 1960s and 70s memory was very expensive (over $1,000,000 per MB), so most systems were random scan. Once memory became cheap, 99% of graphics systems became raster scan.

Resolution (dpi): dots per inch. The higher the dot density, the higher the resolution of the print or scan.

04 Input and hard-copy devices

Input devices

Keyboards, button boxes, dials
Mouse
Trackballs and space balls
Joysticks
Data gloves, digitizers, scanners
Touch panels, light pens
Voice systems

Hard-copy devices

Laser printer
Ink-jet printer
Film recorder
Electrostatic printer
Pen plotter

Asked in the midterm

List four hard-copy devices.
Which of the following is an input device? Monitor / Printer / Mouse / Speaker → **Mouse**

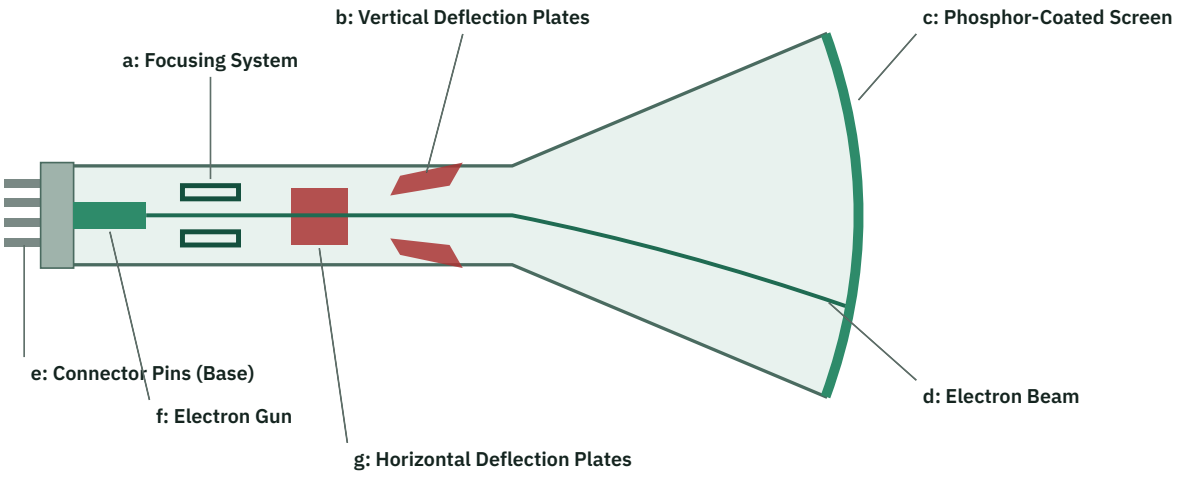
05 Display devices

Video display devices:
Cathode Ray Tube (CRT): Refresh CRT · Direct-view Storage Tube (DVST) · Color CRT monitors
Flat-panel displays: Liquid Crystal Display (LCD) · Plasma panels · Thin-film electroluminescent displays

The slide's tree: projection, direct-view (containing CRT and flat panel) and off-screen. Flat panels split into emissive (OLED, PDP, LED) and non-emissive (LCD).

06 The CRT and its parts

The CRT is the standard design of video monitors (the primary output device). The electron gun fires electrons; they pass through the focusing system, then the deflection plates steer them to any spot on the screen. The screen is coated with phosphor, which glows when an electron hits it.



Part	Function
Connector pins / Base	Connect the tube to the circuit
Electron gun	Fires the electrons (contains a heating filament, cathode and control grid)
Focusing system	Focuses the beam to a point on the screen, like a lens
Vertical / Horizontal deflection plates	Move the beam up/down and left/right (or magnetic deflection coils)
Phosphor-coated screen	Glows when the beam hits it
Electron beam	The beam itself

The focusing system acts like a lens with a focal length such that the center of focus is the screen.
The horizontal and vertical deflectors allow the electron beam to be focused on any spot on the screen.
The screen is coated with a special organic compound called **phosphor**.
For color systems there are groups of three phosphors: red, green, and blue.

Repeats

Rays emitted by a cathode ray tube are _____. → **electrons**
Label the cathode ray tube (the same figure, letters a–g).

07 Persistence and the refresh CRT

Persistence: the time it takes the emitted light from the screen to decay to **one-tenth** of its original intensity.

- Low persistence needs a higher refresh rate to avoid flicker; it is useful for animation.
- High persistence suits highly complex, static pictures.
- General-purpose monitors: 10 to 60 microseconds.

Refresh CRT

The beam is directed to the same screen points repeatedly, keeping the phosphor activated.
Refresh rate: the frequency at which a picture is redrawn on the screen.
Resolution: the maximum number of points/pixels that can be displayed (without overlap) on a CRT.
High-resolution systems are often called **high-definition** systems.

Two ways to display color

1. Beam penetration: layers of colored phosphor; the color depends on how far the beam penetrates (slow beam = red, fast beam = green).
2. Shadow mask: three electron guns (R, G, B) and a mask with holes, each aligned with a triangle of red, green and blue phosphor dots.

08 Resolution and aspect ratio

- In practice, resolution = horizontal points × vertical points, e.g. 800 × 600. A typical high-quality CRT resolution is 1280 × 1024.
- Precisely: points per centimeter.
- Spacing between adjacent pixel centers to avoid overlap: 0.31 mm.

Aspect ratio: the ratio of vertical points to horizontal points necessary to produce equal-length lines in both directions on the screen.

$$AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$$

`Yp = vertical points` `Ymax = vertical height`

`Xp = horizontal points` `Xmax = horizontal width`

That is, vertical point density divided by horizontal point density.

Slide examples

Screen 5 cm × 4 cm, 100 × 100 pixels:

x resolution: $\frac{100}{5} = 20$ pixels per cm

y resolution: $\frac{100}{4} = 25$ pixels per cm

$AR = \frac{25}{20} = \frac{5}{4}$

12-inch screen, width 10 in, height 7 in, resolution 640 × 350:

$AR = \frac{\frac{350}{7}}{\frac{640}{10}} = \frac{50}{64} = 0.78125$

Why use aspect ratio? (repeats)

Aspect ratios are useful for maintaining geometric features of figures drawn on a screen (e.g. a square remains a square and does not become a rectangle, and a circle not an ellipse).

Common ratios: 1:1, 4:3, 16:9, 2.35:1 (width : height).

09 Flat-panel displays

Thinner, less weight, need less power, do not emit radiation. The most common is the LCD. Uses: small TV monitors, calculators, pocket game screens, laptops, airline armrest screens, elevator advertisement boards, rugged portable monitors.

Emissive (emitters)

Convert electrical energy into light.

Plasma panels, thin-film electroluminescent displays, LEDs

Nonemissive (nonemitters)

Use optical effects to convert sunlight or another light source into graphics patterns.

Most important example: LCD

A flat panel is two glass plates with a substance between them. Panels differ in (A) the substance between the plates (LCD or gas plasma) and (B) the arrangement of the transistors (active or passive matrix).

LCD

- Liquid crystal: its molecules have a crystalline arrangement yet flow like a liquid.
- Light, usually a backlight, is blocked or let through depending on how the molecules line up.
- The intersection of two conductors defines a pixel position.
- On: polarized light is twisted so it passes through the opposite polarizer. Off: a voltage aligns the molecules so the light is not twisted.

Plasma panel (gas-discharge)

- Like a neon bulb: a gas mixture (neon) between horizontal and vertical conductors.
- Firing voltage at an intersection turns the gas there into glowing plasma.
- The picture is stored in a refresh buffer and refreshed 60 times per second.
- Its resolution is much better than LCD but it is more expensive, so LCD is used more often.

Active-matrix	Passive-matrix
Each pixel has its own transistor	One transistor controls a whole row or column
Brighter and sharper	Cheaper
More complicated, more expensive	Needs less power

10 Screen clarity

Depends on three qualities: resolution, dot pitch and refresh rate.

Quality	Meaning
Resolution	Number of pixels (H × V). More pixels = clearer. Each pixel gets a color or shade of gray.
Dot pitch	The space between the centers of adjacent pixels. Closer dots = crisper image; it should be less than 0.31 mm.
Refresh rate	How many times the pixels are recharged so their glow remains; usually 45 to 100 times per second.

11 Video card and graphics software

- Video display adapter: a circuit board that determines resolution, number of colors and display speed, with its own memory. 256 KB gives 16 colors; 1 GB gives 16.7 million. Often built into the motherboard.

Type	Meaning
Special purpose packages	For nonprogrammers who want pictures, graphs or charts without worrying about graphics procedures.
General programming packages	A library of graphics functions used from a language such as C, C++ or Java. Example: GL, OpenGL.

The coordinate sequence: modeling → world → viewing and projection → normalized → device coordinates.

Graphics functions

Functions	Role
Output primitives	Building blocks: points, lines, curves, filled areas, character strings
Attributes	How a primitive is displayed: color, line style, text style, fill pattern
Geometric transformations	Change size, position or orientation
Viewing transformations	Select the view, projection type and display location
Input functions	Control data from input devices
Control operations	Housekeeping such as clearing the screen to a color